Chemotherapy Agents *Prodigy Anesthesia*

Below is a table of selected chemotherapy agents and the side effects that have unique anesthetic implications. Note that virtually all chemotherapeutic agents cause nausea and vomiting. Most of them also produce symptoms such as myelosuppression. The side effects more common to chemotherapy agents as a whole have not been listed here.

Drug Name	Class	Therapeutic Uses	Relevant Side Effects
Cyclophosphamide	Nitrogen Mustard	Breast, ovarian, testicular, and lung cancer, Leukemia, lymphoma, myeloma, sarcoma, neuroblastoma, Wilms tumor	Plasma cholinesterase inhibition Hemorrhagic cystitis
Fluorouracil	Pyrimidine analogues	Breast, gastric, lung, and head and neck cancer, acute lymphocytic leukemia, Hodgkin disease, Non-Hodgkin lymphoma	Chest pain
Methotrexate	Folate analogues	breast, lung, and head and neck cancer, acute lymphocytic leukemia, osteogenic sarcoma	Renal dysfunction Hepatic dysfunction
Bleomycin		Testicular, head, and neck cancer, Hodgkin disease, Non-Hodgkin lymphoma	Pulmonary fibrosis
Busulfan	Alkyl sulfonates	Acute and chronic myelogenous leukemia	Thrombocytopenia
Mitomycin	Bioreductive Alkylating Agent	Breast, lung, gastric, colon, rectal, and cervical cancer	Hemolytic uremic syndrome Cardiac failure
Doxorubicin	Anthracyclines	Thyroid, breast, lung, and gastric cancer, Hodgkin disease, non-Hodgkin lymphoma, acute lymphocytic leukemia, neuroblastoma	Cardiomyopathy
Cisplatin	Platinum compounds	Testicular, thyroid, lung, ovarian, endometrial, cervical, and head and neck cancer, neuroblastoma, osteogenic sarcoma	Ototoxicity Renal failure Peripheral neuropathy
Paclitaxel	Taxanes	Breast, lung, bladder, and ovarian cancer	Hypersensitivity reactions Peripheral neuropathy
Vincristine	Vinca alkaloids	Hodgkin disease, Non-Hodgkin lymphoma, Wilms tumor, neuroblastoma, small cell lung cancer	Peripheral and autonomic neuropathy
Tamoxifen	Antiestrogens	Breast cancer	Venous thrombosis Hypercalcemia

Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 553-555.
Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 501.

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